# Freight Cargo Inventory and Shipment Management System Project Progress Report 2 - RM

Muhammad Ahmad 30239993

Syed Essam Uddin Khawaja 30242943

Muhammad Zohaib Talat 30257908

CPSC471 W26

Group G-4

**Assumptions:**

1: User.user_id uniquely identifies each user and email.

2: Every user has exactly one role through role_id.

3: A user is either an Admin or a WarehouseStaff, never both.

4: Each warehouse is managed by exactly one Admin.

5: An admin may manage zero or more Warehouses.

6: Each Warehouse is located at exactly one Location.

7. Each InventoryRecord represents one CargoItem stored in one Warehouse.

8. (warehouse_id, cargo_id) uniquely identifies an InventoryRecord.

9. quantity_stored is non-negative.

10. Each Shipment is handled by exactly one Carrier.

11. Each Shipment uses exactly one Route.

12. Each Shipment has exactly one origin and one destination Location.

13. A Carrier may handle zero or more Shipments.

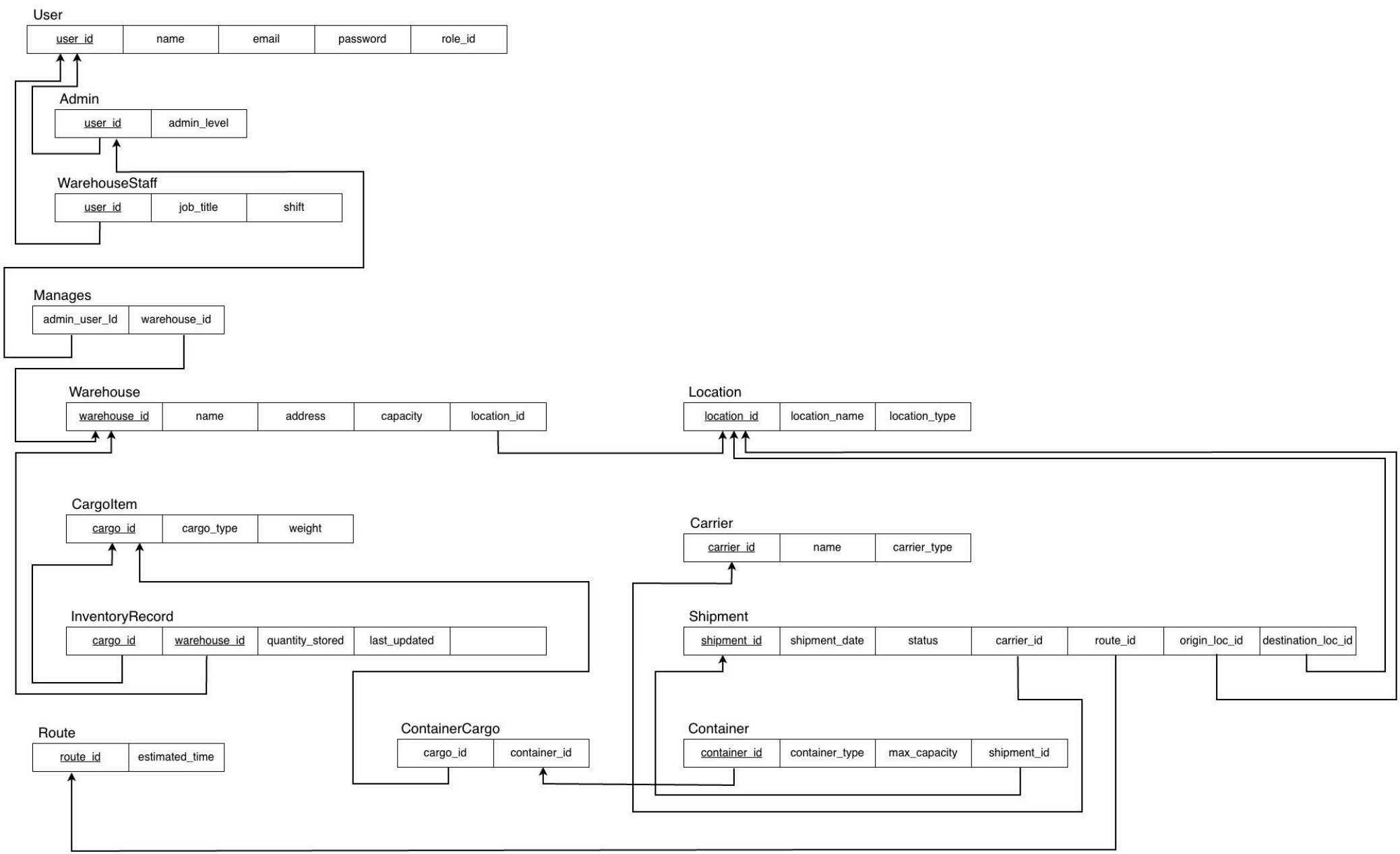
14. A Container may be assigned to at most one Shipment at a time.

15. A Shipment may have one or more Containers.

16. A CargoItem may be stored in at most one Container at a time.

17. ContainerCargo represents the many-to-many relationship between CargoItem and Container.

18. The system stores only the current container–shipment and cargo–container assignments (no history).


User
user_id
name
email
password
role_id
Admin
user_id
admin_level
WarehouseStaff
user_id
job_title
shift
Manages
admin_user_Id
warehouse_id
Warehouse
warehouse_id
name
address
capacity
location_id
Location
location_id
location_name
location_type
CargoItem
cargo_id
cargo_type
weight
Carrier
carrier_id
name
carrier_type
InventoryRecord
cargo_id
warehouse_id
quantity_stored
last_updated
Shipment
shipment_id
shipment_date
status
carrier_id
route_id
origin_loc_id
destination_loc_id
Route
route_id
estimated_time
ContainerCargo
cargo_id
container_id
Container
container_id
container_type
max_capacity
shipment_id